

Slide PTS

PERTEMUAN 7 — SEMESTER 2

PTS — Penilaian Tengah Semester

Informatika - Fase E / Kelas X

SMA Negeri 6 Cimahi

Aturan Ujian

Waktu	70 menit total
PG	20 soal — 40 menit
Esai	5 soal — 30 menit
Nilai	PG 60 + Esai 40 = 100

Materi Ujian

Pert	Materi	TP
1	Array	BK 1.1

Pert	Materi	TP
2	Stack	BK 1.1
3	Queue	BK 1.1
4	Sequential Search	BK 1.3
5	Binary Search	BK 1.3
6	Bubble & Insertion Sort	BK 1.3

Ingat!

Topik	Inti
Array	Indeks mulai 0 , akses acak
Stack	LIFO — push/pop
Queue	FIFO — enqueue/dequeue
Sequential	Cek satu per satu — O(n)
Binary	Data terurut — O(log n)
Bubble	Tukar pasangan — naik ke akhir
Insertion	Sisip ke posisi tepat

Review — Array

A = [10, 20, 30, 40, 50]

- Indeks: 0, 1, 2, 3, 4
- A[2] = 30

- Array 2D: B[2][3]

Review — Stack (LIFO)

```
push(5) → [5]
push(3) → [5, 3]
pop() → [5] (3 keluar)
push(7) → [5, 7]
```

Seperti tumpukan piring!

Review — Queue (FIFO)

```
enqueue(2) → [2]
enqueue(4) → [2, 4]
dequeue() → [4] (2 keluar)
enqueue(6) → [4, 6]
```

Seperti antrean kasir!

Review — Sequential Search

[15, 8, 23, 42, 10, 31, 5, 18]

Mencari **31** > 15, 8, 23, 42, 10, **31**

— 6 perbandingan

Review — Binary Search

Untuk data **TERURUT!**

[3, 7, 10, 15, 20, 25, 30, 35, 42, 50]

Mencari **25**: mid = 4 (20) → kanan → mid = 7 (35) → kiri → mid = 5 (**25**)

Hanya **3 langkah!** (Sequential butuh 6)

Review — Bubble Sort

[42, 15, 8, 23, 10]

```
Pas 1: [15, 8, 23, 10, 42] → 42 mengapung  
Pas 2: [8, 15, 10, 23, 42] → 23 mengapung  
Pas 3: [8, 10, 15, 23, 42] → urut!
```

4 pas, 10 pertukaran

Review — Insertion Sort

[42, 15, 8, 23, 10]

```
Ambil 42 → [42]  
Ambil 15 → [15, 42]  
Ambil 8 → [8, 15, 42]  
Ambil 23 → [8, 15, 23, 42]  
Ambil 10 → [8, 10, 15, 23, 42]
```

Lebih efisien untuk data hampir urut

Kerjakan dengan Percaya Diri!

1. Baca soal dengan teliti
2. Dahulukan yang mudah
3. Untuk esai: tulis **langkah-langkahnya**
4. Cek kembali sebelum kumpul

Semoga Sukses!

Setelah PTS

Pert	Materi
8	Pseudocode & Flowchart
9	Input/Output & Kondisi
10	Perulangan
11	Latihan Soal Algoritma

Terima Kasih

MGMP Informatika SMAN 6 Cimahi